

Airbnb Dynamic Pricing Recommendation Engine

Excel Data Preparation and PivotTable – Explanation

This document explains the Airbnb dataset shown in the uploaded Excel screenshots. The data will later be used for Python regression analysis and a dynamic pricing recommendation engine.

1. Airbnb Dataset

The Excel sheet contains Airbnb listing information. The visible columns include neighbourhood/city information, property type, room type, accommodation capacity, bedrooms, bathrooms, amenities, price, review score and number of reviews. These variables are useful for understanding which factors are associated with Airbnb prices.

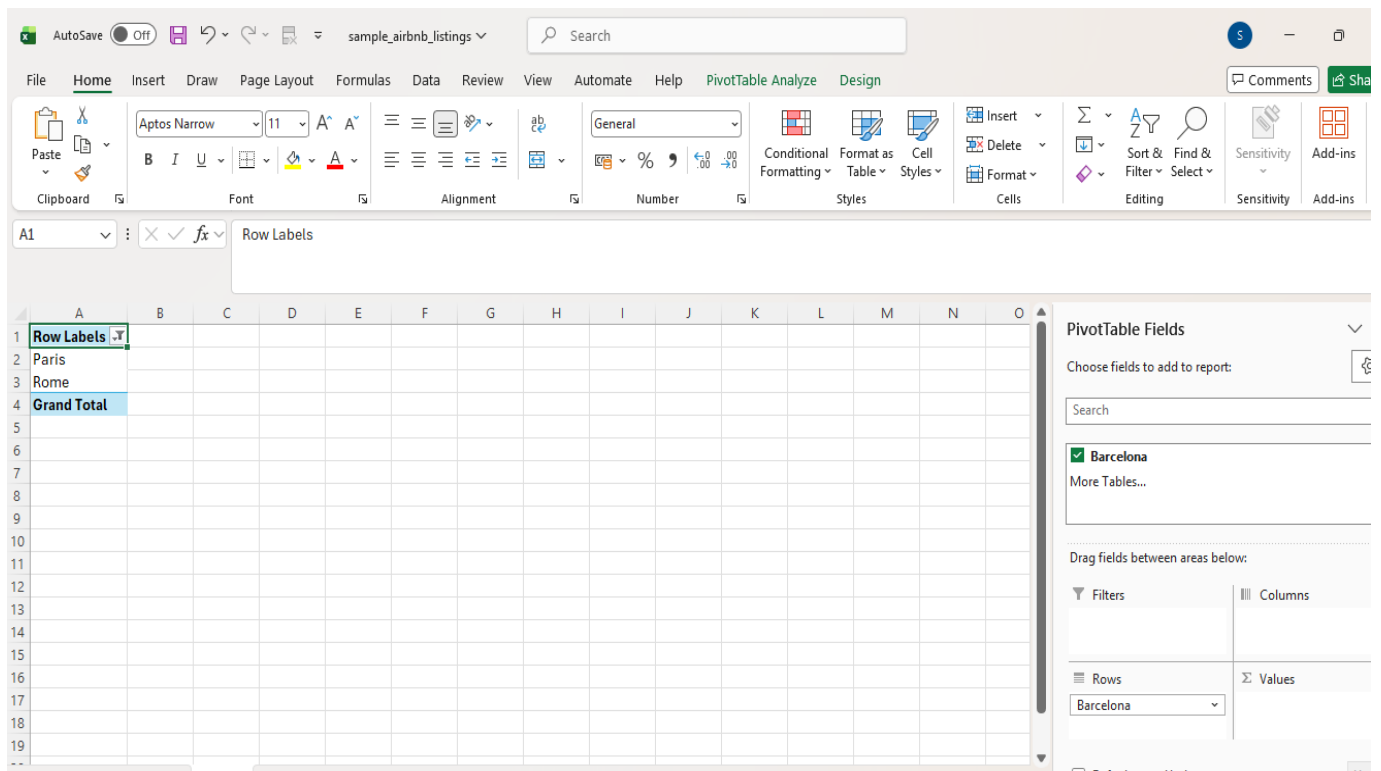


Figure 1: Airbnb listing data in Excel. The visible sample includes values such as price, review score and number of reviews.

2. Important Columns for the Project

Column	Purpose in Pricing Project
City / Location	Used to compare prices across different locations.
Property Type	Used to compare pricing for different property categories.
Room Type	Shows whether the listing is an entire home, private room, etc.
Accommodates	Indicates how many guests the property can accommodate.
Bedrooms	Can influence the property's expected price.
Bathrooms	Can be used as a listing-quality/property-size predictor.
Amenities	Describes facilities available at the property.
Price	Target variable that the regression model will predict.
Review Score	Represents listing quality/customer rating.
Number of Reviews	Shows the amount of guest review history.

3. PivotTable Analysis

The first Excel screenshot shows a PivotTable. A PivotTable is used to summarize a large dataset quickly. In the screenshot, the field 'Barcelona' is placed in the Rows area. This indicates that the data is being grouped by the selected location/category.

D	E	F	G	H	I	J	K	L	
property_type	room_type	accommodates	bedrooms	bathrooms	amenities	price	review_score	number_of_reviews	
entire home	Entire home	12	0	1	Wifi, TV, V	144.09	3.65	158	
entire home	Entire home	9	4	2	Kitchen, A	489.59	4.98	194	
hotel room	Hotel room	8	4	1	Pool, Heat	0	4.4	268	
entire home	Entire home	8	4	1.5	TV, Air con	356.72	3.99	74	
entire home	Entire home	8	4	1.5	Washer, V	378.99	4.49	313	
entire home	Entire home	8	4	2	Kitchen, H	269.5	4.85	137	
entire home	Entire home	8	4	1	Wifi, Kitch	289.78	3.65	242	
entire home	Entire home	8	4	1.5	TV, Air con	356.72	3.99	74	
entire home	Entire home	7	3	1	TV, Wifi, A	287.75	4.01	256	
entire home	Entire home	7	3	1.5	Air conditi	186.33	4.44	150	
entire home	Entire home	7	3	1	Workspac	350.57	3.63	255	
entire home	Entire home	7	3	1.5	Elevator, k	166.1	4.56	101	
entire home	Entire home	6	3	1	TV, Works	221.34	4.96	223	
entire home	Entire home	5	2	1.5	Kitchen, A	230.36	4.47	22	
entire home	Entire home	5	2	1	Washer, P	129.11	4.78	120	
hotel room	Hotel room	5	2	1	Workspac	114.96	4.58	118	
entire home	Entire home	5	2	1.5	Heating, P	243.4	4.02	133	
entire home	Entire home	4	2	1	TV, Pool, h	160.87	4.47	254	
entire home	Entire home	4			Workspac	160.43	4.68	66	
entire home	Entire home	4	2	1	Elevator, V	151.24	3.89	270	

Figure 2: PivotTable Fields panel and Rows area. This is the starting point for summarizing Airbnb listings.

4. How the PivotTable Helps the Project

1. Select the Airbnb data range.
2. Insert a PivotTable from the Insert menu.
3. Place City/Location in the Rows area.
4. Place Price in the Values area and set it to Average.
5. Add Property Type, Room Type or Season as filters when required.
6. Use the resulting summary to compare average Airbnb prices.

5. Example Airbnb Records

	D	E	F	G	H	I	J	K	L
1									
2	property_type	room_type	accommodations	bedrooms	bathrooms	amenities	price	review_score	number_of_reviews
3	Entire home/apt	Entire home/apt	12	0	1	Wifi, TV, W	144.09	3.65	158
62									
63									
64									
65									
66									
67									
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Figure 3: Sample Airbnb records. The screenshot shows different property/room types, accommodation values, bedrooms, bathrooms, amenities, prices, review scores and review counts.

6. Connection to the Python Pricing Engine

After the Excel analysis, the same dataset can be loaded into Python. Python will clean the data, analyze price patterns, train a Linear Regression model, evaluate prediction accuracy and generate a recommended price. The predicted-price output can then be exported as a CSV file and imported into Tableau.

Excel Dataset	→	Data Cleaning
Data Cleaning	→	Regression Model
Regression Model	→	Predicted Price
Predicted Price	→	Dynamic Price Adjustment



7. Suggested Project Conclusion

The Excel analysis provides an initial understanding of Airbnb pricing patterns by location, property type, reviews and listing characteristics. These findings can be combined with a Python regression model to estimate a suitable listing price. The final predictions can be visualized in Tableau using filters for location, property type, room type and other relevant factors.